SAT Math Mastery · Session 7

Data Analysis and Statistics

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	ratios, rates and unit conversion · mean, median and mode from tables · how adding, removing or changing values changes the mean and standard deviation · box plots and spread · sampling and study design · margin of error and estimates · probability and two-way tables · scatterplots, slope and residuals
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos (or a calculator), one expression per line; tap a grey dot to read an intersection or a vertex.

1. HARD · DESMOS

Study room A has x students and study room B has 72 students. One-third of the students in room A move to room B. Then one-fourth of the students who are now in room B move to room A. After both moves, the ratio of the number of students in room A to the number in room B is 5 to 7. How many students were in room A originally?

(A) 72 **(B) 36 ✓**

(C) 45 (D) 63

1. After the first move room A has $\frac{2x}{3}$ students and room B has $72+\frac{x}{3}$. After the second move room A has $\frac{3x}{4}+18$ and room B has $54+\frac{x}{4}$.
2. A ratio of $5:7$ means $7A=5B$. Type `y=7(3x/4+18)` and `y=5(54+x/4)`.
3. Tap the intersection: (36, 315). Its x-value is the original number in room A.
4. Check with $x=36$: room A ends with 45 and room B with 63, and $45:63$ is $5:7$.

Answer: (B)

2. HARD · DESMOS

A tram increases its speed at a constant rate from 54 kilometers per hour to 90 kilometers per hour in 20 seconds. What is the tram's acceleration, in meters per minute squared?

1. Speeds in meters per second: Type `54*1000/3600`: it returns 15 and Type `90*1000/3600`: it returns 25.
2. Acceleration (change in speed over time): Type `(25-15)/20`: it returns 0.5 meters per second squared.
3. One minute is 60 seconds, so multiply by 60^2: Type `0.5*60^2`: it returns 1,800.
4. The acceleration is 1800 meters per minute squared.

Answer: 1800

3. Hard · Desmos

A car uses fuel at a rate of 6.0 liters per 100 kilometers on a highway and 10.0 liters per 100 kilometers in the city. On one trip the car travels 240 kilometers on the highway and 80 kilometers in the city. Using 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters, which of the following is closest to the car's fuel efficiency for the whole trip, in miles per gallon?

(A) 87

(B) 6

(C) 30

(D) 34 ✓

1. Fuel used: the highway part uses $240 \times \frac{6.0}{100}$ liters and the city part uses $80 \times \frac{10.0}{100}$ liters. Type `240*6/100+80*10/100`: it returns 22.4 liters.
2. Distance: $240 + 80 = 320$ kilometers. Type `(240+80)/1.6`: it returns 200 miles. Type `22.4/3.8`: it returns 5.8947 gallons.
3. Type `200/(22.4/3.8)`: it returns 33.9286, which is closest to 34 miles per gallon.

Answer: (D)

4. Hard · Desmos

On a map, 1 inch represents 1,500 feet. A rectangular parking lot appears on the map as 2.4 inches by 1.6 inches. Which of the following is the actual area of the lot, in square yards? (1 yard = 3 feet)

(A) 8,640,000

(B) 320,000

(C) 960,000 ✓

(D) 2,880,000

1. Real lengths: Type `2.4*1500`: it returns 3,600 feet and Type `1.6*1500`: it returns 2,400 feet.
2. Change to yards before multiplying: Type `3600/3`: it returns 1,200 yards and Type `2400/3`: it returns 800 yards.
3. Type `1200*800`: it returns 960,000 square yards. (Multiplying in feet first gives square feet, 8,640,000, which is not what was asked.)

Answer: (C)

5. Hard · Desmos

A cyclist rides a 90-kilometer route. She covers the first 60% of the distance at an average speed of 27 kilometers per hour, rests for 30 minutes, and then covers the rest of the route at an average speed of 24 kilometers per hour. What was her average speed for the whole trip, including the rest, in meters per second?

(A) 22.5

(B) 6.25 ✓

(C) 7.08

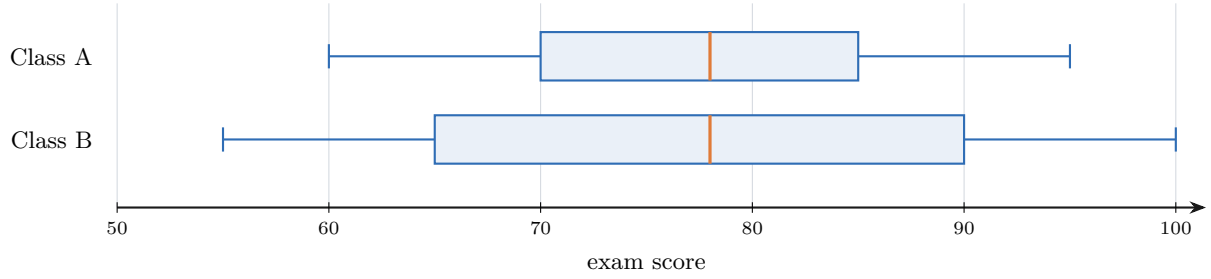
(D) 7.14

1. First part: Type `0.6*90`: it returns 54 km at 27 km/h, so Type `54/27`: it returns 2 hours.
2. Rest of the route: Type `90-54`: it returns 36 km at 24 km/h, so Type `36/24`: it returns 1.5 hours. Total time with the rest: Type `2+1.5+0.5`: it returns 4 hours.
3. Average speed: Type `90/4`: it returns 22.5 km/h. Convert to meters per second: Type `22.5*1000/3600`: it returns 6.25.

Answer: (B)

6. HARD · DESMOS

Box plots summarize final exam scores for two sections of a course.
Class A: minimum 60, $Q1 = 70$, median $= 78$, $Q3 = 85$, maximum 95.
Class B: minimum 55, $Q1 = 65$, median $= 78$, $Q3 = 90$, maximum 100.



Which statement is best supported by these box plots?

(A) The two classes have the same median and the same interquartile range (IQR).

(B) The two classes have the same median, but Class B's scores are more spread out, as shown by its greater IQR. ✓

(C) Class A has a greater median than Class B.

(D) Class B's scores are less spread out than Class A's, as shown by its smaller IQR.

1. Class A: Type `85-70` : it returns 15 is its IQR. Class B: Type `90-65` : it returns 25 is its IQR.
2. Difference of the medians: Type `78-78` : it returns 0, so the medians are equal.
3. Compare the two IQRs: $25 > 15$, so Class B's middle 50% of scores is more spread out.
4. The statement with the same median and a greater IQR for Class B is the one the box plots support.

Answer: (B)

7. HARD · DESMOS

A data set of 24 different numbers has a mean of 50 and a median of 50. A new data set is created by adding 5 to each number in the original data set that is greater than the median and subtracting 5 from each number that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

(A) Median

(B) Mean

(C) Sum of the numbers

(D) Standard deviation ✓

1. Sum change: 12 numbers gain 5 and 12 lose 5. Type `12*5-12*5` : it returns 0, so the sum and the mean do not change.
2. Try a small example with the same pattern: `L=[40,48,52,60]` has mean 50 and median 50. Add 5 to the two numbers above 50 and subtract 5 from the two below: `M=[35,43,57,65]` .
3. Mean and median: Type `mean(L)` : it returns 50, Type `mean(M)` : it returns 50, Type `median(L)` : it returns 50, Type `median(M)` : it returns 50.
4. Standard deviation: Type `stdevp(L)` : it returns 7.2111 and Type `stdevp(M)` : it returns 11.7047. It grew, so the standard deviation is the measure that changes.

Answer: (D)

8. HARD · DESMOS

The mean score of 12 students on a quiz was 16.5 points. If the highest individual score is removed, the mean score of the remaining 11 students becomes 15 points. What was the highest score?

1. Total of all 12 scores: Type `12*16.5` : it returns 198.
2. Total of the other 11 scores: Type `11*15` : it returns 165.
3. The highest score is the difference: Type `198-165` : it returns 33.
4. The highest score is 33.

Answer: 33

9. HARD · DESMOS

A weather station recorded the high temperature, in degrees Fahrenheit, on each of 15 days. The mean of these 15 readings was 42. After the reading from one day was removed as a recording error, the mean of the remaining 14 readings was 40. What was the removed reading?

1. Total of the 15 readings: Type `15*42` : it returns 630.
2. Total of the other 14 readings: Type `14*40` : it returns 560.
3. The removed reading is the difference: Type `630-560` : it returns 70.
4. The removed reading is 70 degrees.

Answer: 70

10. HARD · DESMOS

A data set is 10, 14, 18, 22, 36. A new value of 18 is added to the set. Which of the following statements about the new data set is (are) true?
I. The mean is less than the mean of the original set. II. The median is the same as the median of the original set. III. The range is the same as the range of the original set.

(A) I only

(B) I and II only

(C) II and III only

(D) I, II and III ✓

1. Type the original set and the new set: `L=[10,14,18,22,36]` and `M=[10,14,18,18,22,36]` .
2. Original: Type `mean(L)` : it returns 20, Type `median(L)` : it returns 18, Type `max(L)-min(L)` : it returns 26 (range).
3. New: Type `mean(M)` : it returns 19.6667, Type `median(M)` : it returns 18, Type `max(M)-min(M)` : it returns 26 (range).
4. The mean is smaller ($19.6667 < 20$), the median is the same (18) and the range is the same (26): all three statements are true.

Answer: (D)

11. Hard · Desmos

A group of 30 people has a mean age of 70 years. When a second group of 20 people is added, the mean age of all 50 people is 76 years. What is the mean age, in years, of the second group?

1. Total age of all 50 people: Type `50*76` : it returns 3,800. Total age of the first group: Type `30*70` : it returns 2,100.

2. Second group total: Type `3800-2100` : it returns 1,700.

3. Its mean age: Type `1700/20` : it returns 85.

4. The mean age of the second group is 85 years.

Answer: 85

12. Hard · Desmos

The frequency table shows the scores of 30 students on a 5-point quiz.

Score	1	2	3	4	5
Frequency	3	5	8	9	5

What is the median score?

(A) 2.5

(B) 3 ✓

(C) 3.5

(D) 4

1. Type the 30 scores in two lists: `A=[1,1,1,2,2,2,2,2]` and `B=[3,3,3,3,3,3,3,3,4,4,4,4,4,4,4,4,5,5,5,5,5]` .

2. Check the count: Type `length(join(A,B))` : it returns 30.

3. Median: Type `median(join(A,B))` : it returns 3.

4. The median score is 3.

Answer: (B)

13. Hard · Desmos

A data set has eight values: 12, 15, 18, 20, 20, 22, 25, 28. The values 12 and 28 are removed and the values 19 and 21 are added in their place. Which of the following describes the mean and the standard deviation (SD) of the new data set compared with the original data set?

(A) Mean smaller, SD smaller

(B) Mean same, SD larger

(C) Mean same, SD smaller ✓

(D) Mean same, SD same

1. Type the original and the new data set: `L=[12,15,18,20,20,22,25,28]` and `M=[15,18,19,20,20,21,22,25]` .

2. Means: Type `mean(L)` : it returns 20 and Type `mean(M)` : it returns 20.

3. Standard deviations: Type `stdevp(L)` : it returns 4.8218 and Type `stdevp(M)` : it returns 2.7386.

4. The mean is the same and the standard deviation is smaller.

Answer: (C)

14. HARD · DESMOS

A data set consists of the five values 5, 8, 17, x, x, where x is a real number. The variance of a data set is the mean of the squared deviations of its values from their mean. What is the least possible value of the variance of this data set?

(A) 26 (B) 78

(C) 15.6 ✓ (D) 10

1. The mean is $\frac{30+2x}{5}$, so the variance is the average of the five squared distances from it.

2. Type `y=((5-(30+2x)/5)^2+(8-(30+2x)/5)^2+(17-(30+2x)/5)^2+2(x-(30+2x)/5)^2)/5`.

3. Tap the lowest point of the curve: (10, 15.6). The least possible variance is the y-value.

4. Check with $x = 10$ (mean 10): Type `((5-10)^2+(8-10)^2+(17-10)^2)/5`: it returns 15.6.

Answer: (C)

15. HARD · DESMOS

A data set has 10 values: 31, 38, 42, 45, 47, 52, 53, 58, 65, 69. The mean of the data set is 50. The values 47 and 53 are removed, leaving 8 values. Which of the following is true of the new data set compared with the original? (SD means standard deviation.)

(A) Mean greater; SD smaller (B) Mean unchanged; SD smaller

(C) Mean unchanged; SD larger ✓ (D) Mean smaller; SD larger

1. Type the original and the new data set: `L=[31,38,42,45,47,52,53,58,65,69]` and `M=[31,38,42,45,52,58,65,69]`.

2. Means: Type `mean(L)`: it returns 50 and Type `mean(M)`: it returns 50.

3. Standard deviations: Type `stdevp(L)`: it returns 11.2517 and Type `stdevp(M)`: it returns 12.49.

4. The mean is unchanged and the standard deviation is larger.

Answer: (C)

16. HARD · DESMOS

A list of nine positive integers has a mean of 12 and a median of 9. What is the greatest possible value of the sum of the three largest integers in the list?

1. Total of the nine integers: Type `9*12`: it returns 108.

2. Make everything except the three largest as small as possible: four 1s, the median 9 and a sixth integer 9. Then Type `108-4*1-9-9`: it returns 86.

3. Check with an example: `L=[1,1,1,1,9,9,9,9,68]`. Type `mean(L)`: it returns 12 and Type `median(L)`: it returns 9.

4. The greatest possible sum of the three largest integers is 86.

Answer: 86

17. HARD · DESMOS

Seven positive integers, not necessarily distinct, have a sum of 101 and a median of 11. What is the greatest possible value of the mean of the three largest of these integers?

(A) 29 ✓ (B) 31.67

(C) 32.33 (D) 33.67

1. Everything except the three largest is as small as possible: three 1s and the median 11. Type `101-3*1-11`: it returns 87.
2. Mean of the three largest: Type `87/3`: it returns 29.
3. Check with an example: `L=[1,1,1,11,11,11,65]`. Type `total(L)`: it returns 101 and Type `median(L)`: it returns 11.
4. The greatest possible mean is 29.

Answer: (A)

18. HARD · DESMOS

A data set has six values: 2, 3, 5, 12, 18, 20. Every value is increased by 4, and then a seventh value equal to the mean of the six increased values is added to the data set. Consider the following statements about the new data set of seven values compared with the original data set.
I. The mean is 4 greater.
II. The standard deviation is smaller.
III. The median is 4 greater.
Which of the statements must be true?

(A) I, II and III (B) I only

(C) I and II only ✓ (D) II and III only

1. Type the original data and the new data (each value plus 4, with the mean 14 added): `L=[2,3,5,12,18,20]` and `M=[6,7,9,14,16,22,24]`.
2. Means: Type `mean(L)`: it returns 10 and Type `mean(M)`: it returns 14. The mean is 4 greater: I is true.
3. Standard deviations: Type `stdevp(L)`: it returns 7.1414 and Type `stdevp(M)`: it returns 6.6117. It is smaller: II is true.
4. Medians: Type `median(L)`: it returns 8.5 and Type `median(M)`: it returns 14. The median is 5.5 greater, not 4: III is false.

Answer: (C)

19. HARD · DESMOS

A data set has a median of 52 and an interquartile range (IQR) of 14. Each value x in the data set is replaced by $15 - \frac{x}{2}$. What is the sum of the median and the IQR of the new data set?

(A) 3 (B) 11

(C) -18 **(D) -4 ✓**

1. Median of the new data: Type `15-52/2`: it returns -11.

2. IQR of the new data (halved; the shift of 15 does not matter): Type `14/2`: it returns 7.

3. Sum: Type `-11+7`: it returns -4.

4. The sum is -4.

Answer: (D)

20. HARD · DESMOS

Let a, b, and c be positive integers. If the average (arithmetic mean) of a, b, and c is 100, which of the following is NOT a possible value of any of the three integers?

(A) 150 (B) 250

(C) 298 **(D) 299 ✓**

1. The three integers add to Type `3*100`: it returns 300.

2. The largest possible integer uses 1 and 1 for the other two: Type `300-1-1`: it returns 298.

3. Test 299: Type `299+1+1`: it returns 301, which is more than 300, so it is impossible.

4. The value that is not possible is 299.

Answer: (D)

21. HARD · NO CALCULATOR NEEDED

A researcher wants to determine whether a flipped-classroom method (Method A) causes higher unit-test scores than traditional lecture (Method B) among Grade 11 Algebra II students in a large school district. From all Grade 11 Algebra II sections in the district, the researcher randomly assigns 10 sections to use Method A and 10 sections to use Method B for one unit, then compares the sections' mean test scores. Which statement is best supported by this study design?

(A) Because sections were randomly assigned to a method, a causal conclusion about the effect of the method on test scores can be drawn for the Grade 11 Algebra II sections in this district, but the results should not be generalized beyond this district. ✓

(B) Because sections, rather than individual students, were randomly assigned, no causal conclusion can be drawn at all.

(C) Because an entire school district was involved, the results can be generalized to all high school students nationally.

(D) Because the researcher chose which sections received which method, this is an observational study, so no causal conclusions are valid.

1. The researcher assigned the sections to the two methods at random. Random assignment is what supports a cause-and-effect conclusion about the method for the sections studied.
2. The statements that call it an observational study or say that no causal conclusion can be drawn at all are wrong: random assignment of sections is still random assignment.
3. The sections come from this one district. Nothing was randomly selected from all districts in the country, so the results cannot be generalized nationally or beyond this district.

Answer: (A)

22. HARD · NO CALCULATOR NEEDED

For a research orchard, a scientist randomly selected 40 apple saplings from the saplings that were 3 years old at the orchard and randomly assigned them to two groups. Each group received a different irrigation schedule for 2 consecutive years to observe whether the schedule affects fruit yield. Based on the design of the study, what is the largest group to which these results can be applied?

(A) All 40 saplings that were selected at this orchard

(B) All apple saplings that were 3 years old at this orchard ✓

(C) All apple saplings that were 3 years old in the country

(D) All apple saplings that were 5 years old at this orchard

1. The saplings were randomly selected from the 3-year-old saplings at the orchard, so the results apply to that population.
2. Random assignment to the two irrigation schedules supports a cause-and-effect conclusion for that population.
3. The 5-year-old saplings and the saplings elsewhere in the country were not part of the selection, and the 40 saplings alone are a smaller group than the population. The largest group is all 3-year-old apple saplings at this orchard.

Answer: (B)

23. HARD · DESMOS

At the end of a radio program, the host invited listeners to text in and respond to a poll asking, "Do you support extending school hours by 30 minutes?" By the end of the show, the host reported that 62% responded "Yes" and 35% responded "No." Which of the following best explains why the results are unlikely to represent the sentiments of all students' families in the country?

(A) The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.

(B) Those who responded to the poll were not a random sample of all students' families in the country. ✓

(C) There were not 50% "Yes" responses and 50% "No" responses.

(D) The show did not allow listeners enough time to respond to the poll.

1. Add the reported percentages: Type `62+35`: it returns 97.
2. The missing part: Type `100-97`: it returns 3 percent gave no answer or another answer. That does not make the poll invalid.
3. Listeners chose for themselves whether to text in (a voluntary-response sample), so the sample is not random.
4. The best explanation is that the respondents were not a random sample of all students' families.

Answer: (B)

24. HARD · DESMOS

A polling agency surveyed 800 adults selected at random from a large city and asked, "Are you satisfied with the quality of public transportation in the city?" Of those surveyed, 71% responded that they were satisfied. Based on the results of the survey, which of the following statements must be true?

1. Of all adults in the city, 71% are satisfied with the quality of public transportation.
2. If another 800 adults selected at random from the city were surveyed, 71% of them would report they are satisfied.
3. If 800 adults selected at random from a different city were surveyed, 71% of them would report they are satisfied.

(A) None ✓

(B) 2 only

(C) 1 and 2 only

(D) 1 and 3 only

1. The sample count: Type `0.71*800`: it returns 568 satisfied adults out of 800. Check: Type `568/800`: it returns 0.71.
2. That is a percentage for this sample only, so statement 1 (the whole city) does not have to be true.
3. A new random sample would give a different count, so statement 2 does not have to be true.
4. Another city is a different population, so statement 3 does not have to be true. None must be true.

Answer: (A)

25. HARD · DESMOS

A survey of $n = 400$ randomly selected residents produces a margin of error of 4.5% at a given confidence level. Assuming the margin of error is inversely proportional to $\sqrt{n}$, what would the margin of error be, in percent, for a survey of $n = 1600$ residents at the same confidence level?

1. The sample size is multiplied by Type `1600/400` : it returns 4.

2. The margin of error is divided by the square root of that: Type `sqrt(1600/400)` : it returns 2.

3. New margin of error: Type `4.5/2` : it returns 2.25.

4. Check in one line: Type `4.5*sqrt(400/1600)` : it returns 2.25 percent.

Answer: 2.25

26. HARD · DESMOS

Based on a random sample survey, a researcher estimates that $24\% \pm 4\%$ of high school students in a district spend more than 2 hours a day on e-reading apps. A school administrator claims that 32% of the district's high school students spend more than 2 hours a day on such apps. Is the administrator's claim consistent with the survey's estimate?

(A) Yes, because 32% is within the range of 20% to 28% suggested by the estimate.

(B) Yes, because 32% is close to 24%.

(C) No, because the sample size is too small to draw any conclusion.

(D) No, because 32% is outside the range of 20% to 28% suggested by the estimate. ✓

1. Lower end: Type `24-4` : it returns 20. Upper end: Type `24+4` : it returns 28.

2. The plausible range is 20% to 28%.

3. Compare: Type `32-28` : it returns 4 percentage points above the top of the range.

4. So 32% is outside the range, and the claim is not consistent with the survey.

Answer: (D)

27. HARD · DESMOS

Two polls use the same confidence level. For a poll of n people with sample proportion $\hat{p}$, the margin of error is proportional to $\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$. Poll A surveyed 625 people and had $\hat{p} = 0.50$. Poll B had $\hat{p} = 0.10$ and a margin of error half as large as the margin of error of Poll A. How many people did Poll B survey?

1. Poll A: Type `sqrt(0.5*0.5/625)` : it returns 0.02.

2. Poll B has half of that: Type `0.02/2` : it returns 0.01.

3. Solve $\sqrt{\dfrac{0.1(0.9)}{n}} = 0.01$ for n: Type `0.1*0.9/0.01^2` : it returns 900.

4. Poll B surveyed 900 people.

Answer: 900

28. HARD · DESMOS

A researcher surveys a random sample of 625 workers in a city and calculates a 95% confidence interval of 24.2 to 27.8 minutes for the mean one-way commute time of all workers in the city. Assume a second random sample of 5,625 workers has similar variability and the same method is used. Which of the following statements must be true?

(A) About 95% of sampled commutes lie in the interval.

(B) The new margin is about 0.2 min.

(C) The population mean cannot be 28.1 min.

(D) The new margin is about 0.6 min. ✓

1. Margin of error of the first interval: Type `(27.8-24.2)/2`: it returns 1.8.
2. The sample is multiplied by Type `5625/625`: it returns 9.
3. New margin of error: Type `1.8/sqrt(9)`: it returns 0.6.
4. The interval is about the mean of all workers, not individual commutes, and a 95% interval does not rule out other values with certainty. So the new margin of about 0.6 minutes is the true statement.

Answer: (D)

29. HARD · DESMOS

County P has 2,500 adult residents and county Q has 900. A random sample of county P residents found that 52% support a proposed bike lane, with a margin of error of 3 percentage points. A separate random sample of county Q residents found that 38% support it, with a margin of error of 4 percentage points. Based on the margins of error, what is the greatest possible value of (number of adult supporters in county P) − (number of adult supporters in county Q)?

(A) 1,033

(B) 1,069 ✓

(C) 525

(D) 958

1. The most supporters in county P come from the top of its range: $52 + 3 = 55\%$ of 2,500. Type `0.55*2500`: it returns 1,375.
2. The fewest supporters in county Q come from the bottom of its range: $38 - 4 = 34\%$ of 900. Type `0.34*900`: it returns 306.
3. The greatest difference is Type `1375-306`: it returns 1,069.

Answer: (B)

30. HARD · DESMOS

In a career-interest survey, 90 Grade 11 students intend a non-STEM track: 40 of them are male and 50 are female. Two of these students are selected at random, one after another, without replacement. What is the probability that both selected students are female? (Round to the nearest thousandth.)

1. First selection: $\frac{50}{90}$ chance of a female student. With no replacement, 49 of the remaining 89 students are female.
2. Multiply the two probabilities: Type `50/90*49/89`: it returns 0.3059.
3. Rounded to the nearest thousandth: 0.306.

Answer: 0.306

31. HARD · DESMOS

On January 1, 2024, there were 34.1 million mobile broadband subscriptions in a certain country. By January 1, 2025, the number of subscriptions had increased to 36.8 million. If a subscriber on January 1, 2025, is selected at random, the probability that the subscriber uses Carrier A is 0.45. There were p million Carrier A subscribers on January 1, 2025. To the nearest integer, what is the value of p?

1. The 2024 figure is not needed. Use the total on January 1, 2025: 36.8 million.

2. Carrier A has 45% of them: Type `36.8*0.45`: it returns 16.56.

3. To the nearest integer, $p = 17$.

Answer: 17

32. HARD · DESMOS

The table below shows the distribution of employment type and sector for a group of employees at a company.

	Public sector	Private sector
Full-time	45	12
Part-time	9	x

If one of these employees who works in the private sector is chosen at random, the probability that the employee is part-time is $\frac{1}{5}$. What is the value of x?

1. The probability of part-time among the private-sector employees is $\frac{x}{12+x}$.

2. Type `y=x/(12+x)` and `y=1/5`.

3. Tap the intersection: (3, 0.2). Its x-value is the answer.

4. Check: Type `3/(12+3)`: it returns 0.2, which is $\frac{1}{5}$.

Answer: 3

33. HARD · DESMOS

	Online	Paper form
Robotics	60%	80%
Sports	25%	10%
Art	10%	5%
Music	5%	5%

A school activity survey asked each student to select the one activity they preferred for a school event. Some students responded online, and the others responded on a paper form. The table above shows the distribution of preferred activity, in percent, for each response type. For the survey, the number of paper-form responses was twice the number of online responses. If a student who preferred Art is selected at random, what is the probability that the student responded on the paper form? (Express your answer as a decimal or fraction, not as a percent.)

1. Use 100 online responses and 200 paper responses.

2. Art online: Type `0.1*100` : it returns 10. Art on paper: Type `0.05*200` : it returns 10.

3. Probability of paper given Art: Type `10/(10+10)` : it returns 0.5.

4. The probability is $\frac{1}{2}$ (or 0.5).

Answer: $\frac{1}{2}$ **(or** 0.5**)**

34. HARD · DESMOS

A conference has 300 participants. Each participant attended either the Technical track or the Business track, and either in person or online. Some of the data are shown.

	Technical	Business	Total
In person	70		
Online	50		
Total	120		300

Of the participants in the Business track, 35% attended online. How many of the 300 participants attended in person?

(A) 187 ✓ (B) 63

(C) 113 (D) 145

1. Business track: Type `300-120`: it returns 180.
2. Online Business: Type `0.35*180`: it returns 63.
3. In-person Business: Type `180-63`: it returns 117.
4. In person in total: Type `70+117`: it returns 187.

Answer: (A)

35. HARD · DESMOS

A survey of 240 adults asked whether they commute by bicycle and whether they own a helmet. Of the adults who commute by bicycle, $\frac{3}{5}$ own a helmet. Of the adults who do not commute by bicycle, $\frac{1}{4}$ own a helmet. If one adult is selected at random from those who own a helmet, the probability that the selected adult commutes by bicycle is $\frac{12}{19}$. How many of the 240 adults own a helmet?

(A) 60 **(B)** 95 ✓

(C) 144 (D) 100

1. Let b be the number who commute by bicycle. Helmet owners: $\frac{3b}{5}$ among cyclists plus $\frac{240-b}{4}$ among the others.
2. The probability that a helmet owner is a cyclist is $\dfrac{3b/5}{3b/5+(240-b)/4} = \dfrac{12}{19}$. Type `y=(3x/5)/(3x/5+(240-x)/4)` and `y=12/19` (letting x stand for b).
3. Tap the intersection: (100, 0.6316), so $b = 100$.
4. Helmet owners in all: Type `3*100/5+(240-100)/4`: it returns 95.

Answer: (B)

PART B · SELF-ASSESSMENT — 20 PROBLEMS

36. MEDIUM · DESMOS

The table shows the number of siblings reported by 20 students.

Number of siblings	0	1	2	3	4
Number of students	5	8	4	2	1

What is the mean number of siblings for these 20 students?

(A) 1.1 (B) 1.2

(C) 1.3 ✓ (D) 1.5

1. Each number of siblings is counted as many times as there are students: $0 \cdot 5 + 1 \cdot 8 + 2 \cdot 4 + 3 \cdot 2 + 4 \cdot 1$. Type `0*5+1*8+2*4+3*2+4*1` : it returns 26.
2. Divide by the 20 students. Type `26/20` : it returns 1.3.
3. The mean is 1.3 siblings.

Answer: (C)

37. MEDIUM · DESMOS

The table below shows the distribution of ages of 20 tutors enrolled in a training program.

Age	Frequency
20	7
21	5
22	3
23	2
24	1
30	2

Which of the following gives the correct order of the mean, median, and mode of the ages?

(A) mode < median < mean ✓ (B) mode < mean < median

(C) median < mode < mean (D) mean < mode < median

1. Type the 20 ages as a list: `A=[20,20,20,20,20,20,20,21,21,21,21,21,22,22,22,23,23,24,30,30]`.
2. Type `mean(A)` : it returns 22.05. Then Type `median(A)` : it returns 21.
3. The mode is the most frequent age: 20 appears 7 times in the list, more than any other age (Desmos has no mode function, so count).
4. So mode 20 < median 21 < mean 22.05.

Answer: (A)

38. Medium · Desmos

Data set A and data set B each consist of 20 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

Value	Data set A frequency	Data set B frequency
30	2	7
35	5	6
40	6	4
45	7	3

(A) The mean of data set A is greater than the mean of data set B. ✓

(B) The mean of data set A is less than the mean of data set B.

(C) The mean of data set A is equal to the mean of data set B.

(D) There is not enough information to compare the means of the data sets.

1. Type each data set as a list: `A=[30,30,35,35,35,35,35,40,40,40,40,40,40,45,45,45,45,45,45,45]` and `B=[30,30,30,30,30,30,30,35,35,35,35,35,35,40,40,40,40,45,45,45]`.
2. Data set A: Type `mean(A)` : it returns 39.5.
3. Data set B: Type `mean(B)` : it returns 35.75.
4. Since $39.5 > 35.75$, the mean of A is greater.

Answer: (A)

39. Medium · Desmos

The mean of 5 numbers is 14. One of the numbers, 20, is replaced by 8. What is the mean of the new set of 5 numbers? (Enter as a decimal.)

1. Original sum: Type `5*14` : it returns 70.
2. Replace 20 by 8: Type `70-20+8` : it returns 58.
3. New mean: Type `58/5` : it returns 11.6.
4. The new mean is 11.6.

Answer: 11.6

40. Medium · Desmos

If the average (arithmetic mean) of w and x is 40, the average of p and q is 40, and the average of m and n is 40, what is the average of w, x, p, q, m, and n?

1. Each pair adds to Type `40*2` : it returns 80.
2. Three pairs add to Type `3*80` : it returns 240.
3. Mean of the six numbers: Type `240/6` : it returns 40.
4. The average is 40.

Answer: 40

41. Medium · No calculator needed

A researcher surveyed a random sample of chemistry teachers selected from all chemistry teachers in a certain state. The teachers were asked to name the textbook publisher they currently use. What is the largest population to which the results of the survey can be generalized?

(A) All teachers in the country

(B) All chemistry teachers in the country

(C) All chemistry teachers in the state ✓

(D) All teachers in the state

1. Results of a random sample can be generalized only to the population the sample was drawn from.
2. The sample was drawn from chemistry teachers in one state, so the population is all chemistry teachers in that state.
3. Teachers of all subjects, or chemistry teachers in the whole country, were never available to be selected.

Answer: (C)

42. Medium · Desmos

A random sample of 150 residents of a city was surveyed about whether they support a new recycling program. All 150 responded, and 64% said they support it. Based on this information, which of the following statements must be true?
I. Exactly 96 of the 150 residents surveyed said they support the recycling program.
II. Of all city residents, exactly 64% support the recycling program.

(A) Neither

(B) I only ✓

(C) II only

(D) I and II

1. All 150 responded, so 64% of them is a count: Type `0.64*150`: it returns 96 residents.
2. Check: Type `96/150`: it returns 0.64, which is 64%. So statement I must be true.
3. Statement II is about all city residents. A sample percentage is only an estimate of that percentage, so it does not have to be exact.
4. Only statement I must be true.

Answer: (B)

43. Medium · Desmos

A random sample of 500 residents in a county found that 62% support a proposed residential zoning initiative, with a margin of error of 4 percentage points at the 95% confidence level. Which of the following is the most appropriate interpretation of this result?

(A) Exactly 62% of all residents in the county support the initiative.

(B) It is plausible that between 58% and 66% of all residents in the county support the initiative. ✓

(C) Exactly 95% of residents in the county support the initiative.

(D) The survey proves beyond doubt that a majority of residents support the initiative.

1. Lower end of the interval: Type `62-4`: it returns 58.
2. Upper end: Type `62+4`: it returns 66.
3. So it is plausible that between 58% and 66% of all residents support the initiative.
4. The survey gives a plausible range, not an exact value or a proof.

Answer: (B)

44. Medium · No calculator needed

Researchers studying a reforestation program selected a random sample of 250 oak trees planted through the program and measured each tree's growth rate. Based on this sample, which of the following is the most appropriate conclusion?

(A) The average growth rate of the sampled trees can be used to estimate the average growth rate of all oak trees planted through the reforestation program. ✓

(B) The average growth rate of the sampled trees can be used to estimate the average growth rate of all trees, of any species, in the country.

(C) The average growth rate of the sampled trees is exactly equal to the average growth rate of all oak trees planted through the reforestation program.

(D) The average growth rate of the sampled trees cannot be used to estimate the average growth rate of any other trees.

1. Results of a random sample apply to the population it was drawn from: here, the oak trees planted through this program.
2. So the sample average can be used as an estimate for those trees, but it is only an estimate, not an exact value.
3. It cannot speak for trees of every species in the country, and it is not exactly equal to the program's average. Only the statement about estimating the average for the program's oak trees is correct.

Answer: (A)

45. Medium · Desmos

A survey of a random sample of employees at a company estimates that the mean daily commute time is 34 minutes, with an associated margin of error of 4 minutes. Which of the following is the most plausible value for the actual mean commute time, in minutes, of all employees at the company?

(A) 25

(B) 31 ✓

(C) 39

(D) 44

1. Lower end: Type `34-4`: it returns 30. Upper end: Type `34+4`: it returns 38.
2. The plausible values are 30 to 38 minutes.
3. Check the options: $25 < 30$; 39 and 44 are above 38.
4. Only 31 lies in the range.

Answer: (B)

46. Medium · Desmos

A poll of a random sample of residents in a city district estimates that 42% of all residents support a new public transit expansion, with a margin of error of 3.5 percentage points. Which of the following is the most plausible value for the actual percentage of all residents in the district who support the expansion?

(A) 36%

(B) 41% ✓

(C) 47%

(D) 50%

1. Lower end: Type `42-3.5`: it returns 38.5. Upper end: Type `42+3.5`: it returns 45.5.
2. The plausible values are 38.5% to 45.5%.
3. Check the options: 36% is below the range; 47% and 50% are above it.
4. Only 41% lies in the range.

Answer: (B)

47. Medium · Desmos

A survey of a random sample of students at a university estimates that $22\% \pm 3\%$ of the university's 8,000 students use the campus shuttle daily. Based on this estimate, which of the following is closest to a plausible range for the number of students at the university who use the shuttle daily?

(A) $1{,}400 < n < 1{,}800$

(B) $1{,}520 < n < 2{,}000$ ✓

(C) $1{,}760 < n < 1{,}840$

(D) $1{,}600 < n < 2{,}200$

1. The margin of error is 3 points, so the plausible percentages are $22 - 3 = 19$ to $22 + 3 = 25$ percent.

2. Lower end: Type `0.19*8000`: it returns 1,520. Upper end: Type `0.25*8000`: it returns 2,000.

3. The plausible range is $1{,}520 < n < 2{,}000$.

Answer: (B)

48. Medium · Desmos

In a random sample of 60 laptops tested at a distribution center, 15 were found to have a manufacturing defect. Based on this sample, which of the following is the best conclusion about the number of defective laptops in a shipment of 2,400 laptops from the same production run?

(A) Exactly 600 laptops in the shipment are defective.

(B) At least 600 laptops in the shipment are defective.

(C) A best estimate is that 600 laptops in the shipment are defective. ✓

(D) At most 600 laptops in the shipment are defective.

1. Sample defect rate: Type `15/60`: it returns 0.25.

2. Apply the rate to the shipment: Type `0.25*2400`: it returns 600.

3. This comes from a sample, so it is an estimate, not an exact count, a minimum or a maximum.

4. The best conclusion is that about 600 laptops are defective.

Answer: (C)

49. Medium · Desmos

In a random sample of 500 registered voters surveyed about a proposed municipal budget, 310 said they support it. There are 45,000 registered voters in total. Based on the sample, what is the expected number of registered voters who support the proposed budget?

1. Sample proportion in favor: Type `310/500`: it returns 0.62.

2. Apply it to all 45,000 registered voters: Type `0.62*45000`: it returns 27,900.

3. The expected number is 27,900 voters.

Answer: 27,900

50. Medium · Desmos

In a career-interest study, 200 Grade 11 students were surveyed on their intended career track.

	STEM track	Non-STEM track	Total
Male	60	40	100
Female	50	50	100
Total	110	90	200

If a student is selected at random from those surveyed and the student is female, what is the probability that the student intends a STEM track?

(A) 0.45

(B) 0.50 ✓

(C) 0.55

(D) 0.60

1. Number of female students: Type `50+50` : it returns 100.

2. Probability of STEM given female: Type `50/100` : it returns 0.5.

3. Only the female row counts, so the denominator is 100, not 200.

4. The probability is 0.50.

Answer: (B)

51. Medium · Desmos

The table below shows the number of students in two grade levels at a school who do and do not own a graphing calculator.

	Owns a graphing calculator	Does not own a graphing calculator	Total
Grade 10	18	7	25
Grade 11	22	3	25
Total	40	10	50

If one of these 50 students is selected at random, what is the probability that the student is in Grade 11 and does not own a graphing calculator?

(A) $\frac{3}{50}$ ✓

(B) $\frac{7}{50}$

(C) $\frac{10}{50}$

(D) $\frac{22}{50}$

1. Check the total: Type `18+7+22+3` : it returns 50.

2. Grade 11 and no calculator: 3 students. Type `3/50` : it returns 0.06.

3. As a fraction that is $\frac{3}{50}$.

4. The probability is $\frac{3}{50}$.

Answer: (A)

52. MEDIUM · DESMOS

The table below shows the distribution of 50 students by club membership.

	Coding club	Not coding club	Total
Robotics club	15	8	23
Not robotics club	10	17	27
Total	25	25	50

If one of these 50 students is selected at random, what is the probability that the student is in the coding club but not the robotics club? (Express your answer as a decimal or fraction, not as a percent.)

1. Check the total: Type `15+8+10+17`: it returns 50.

2. Coding club but not robotics club: 10 students. Type `10/50`: it returns 0.2.

3. As a fraction that is $\frac{1}{5}$.

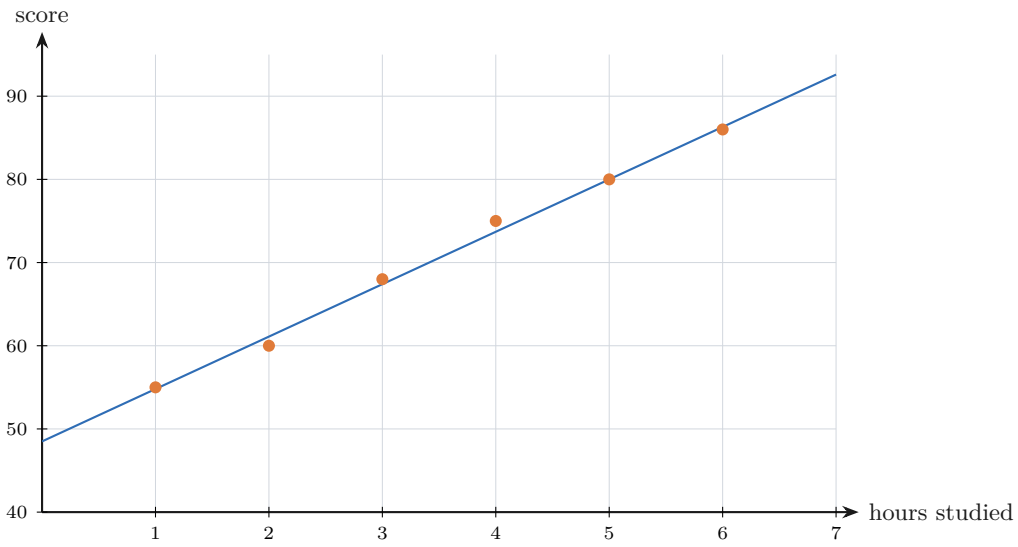
4. The probability is $\frac{1}{5}$ (or 0.2).

Answer: $\frac{1}{5}$ **(or** 0.2**)**

53. Medium · Desmos

A teacher records hours studied (x) and exam score (y) for 6 students.

Hours studied (x)	1	2	3	4	5	6
Exam score (y)	55	60	68	75	80	86



The line of best fit for this data is approximately $y = 6.3x + 48.5$. Which of the following is the best interpretation of the slope of this line?

(A) The predicted exam score for a student who studies 0 hours is 6.3.

(B) Each additional hour studied is associated with a predicted increase of about 6.3 points on the exam. ✓

(C) Each additional point on the exam is associated with a predicted increase of about 6.3 hours studied.

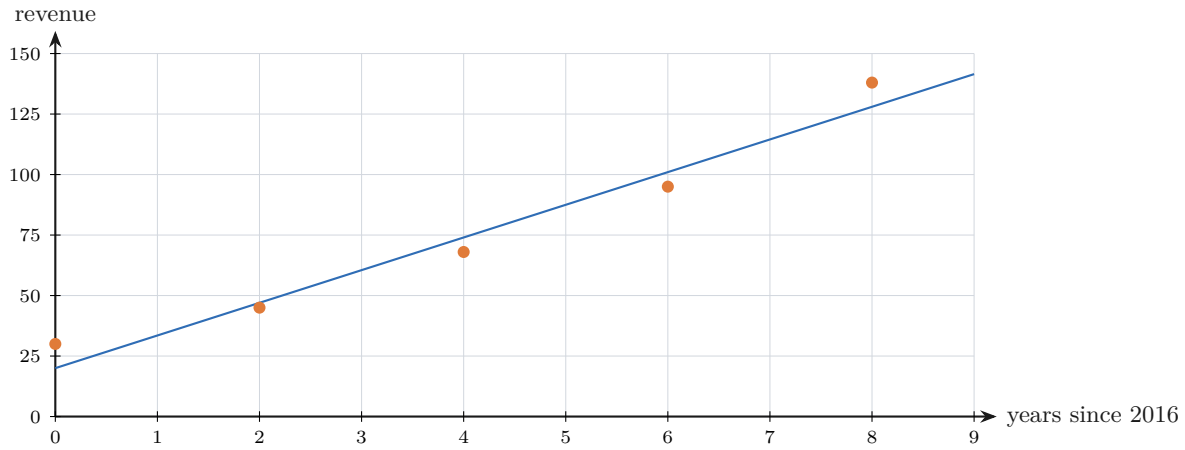
(D) The exam score increases by exactly 6.3 points for every student, regardless of hours studied.

1. Open a table and enter $x_1 = 1, 2, 3, 4, 5, 6$ and $y_1 = 55, 60, 68, 75, 80, 86$.
2. Type `y_1~m*x_1+b`: Desmos returns $m = 6.3429$ and $b = 48.4667$.
3. The slope is about 6.3: the predicted score rises by about 6.3 points for each extra hour studied.
4. The intercept 48.5 is the predicted score at 0 hours, and a best-fit line gives predictions, not exact changes for every student.

Answer: (B)

54. HARD · DESMOS

A company's annual online-sales revenue, in millions of dollars, is recorded for years since 2016 (x). The approximate data points are $(0, 30)$, $(2, 45)$, $(4, 68)$, $(6, 95)$ and $(8, 138)$. The scatterplot shows the data and a line used to model it, $y = 13.5x + 20$.



What is the residual for the data point at $x = 8$?

(A) -10

(B) 0

(C) 10 ✓

(D) 128

1. Predicted value at $x = 8$: Type `13.5*8+20`: it returns 128.

2. Residual = actual − predicted: Type `138-128`: it returns 10.

3. The point is above the line, so the residual is positive (-10 comes from subtracting in the wrong order).

4. The residual is 10.

Answer: (C)

55. HARD · DESMOS

A scatterplot showing the relationship between daily study time (x, in hours) and nightly sleep time (y, in hours) for a group of students has a correlation coefficient of $r = -0.82$. Which of the following best describes this relationship?

(A) A strong, positive linear relationship between study time and sleep time.

(B) A strong, negative linear relationship between study time and sleep time. ✓

(C) A weak, negative linear relationship between study time and sleep time.

(D) No linear relationship between study time and sleep time.

1. The sign of r is negative, so the relationship is negative.

2. The size of r: Type `abs(-0.82)`: it returns 0.82. Check how far it is from 1: Type `1-0.82`: it returns 0.18.

3. It is close to 1, so the relationship is strong.

4. It is a strong, negative linear relationship.

Answer: (B)